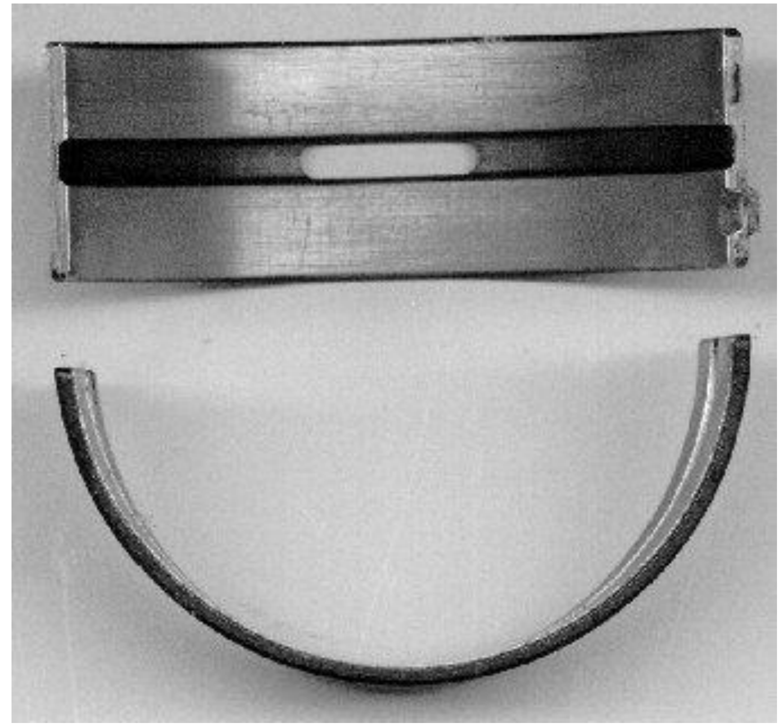
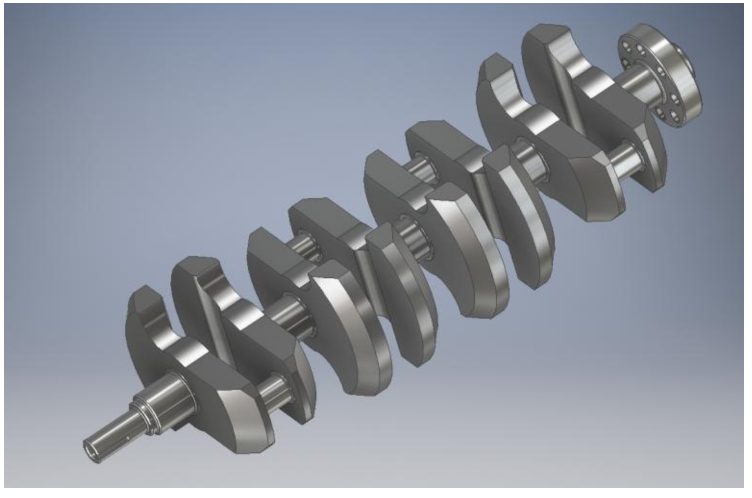
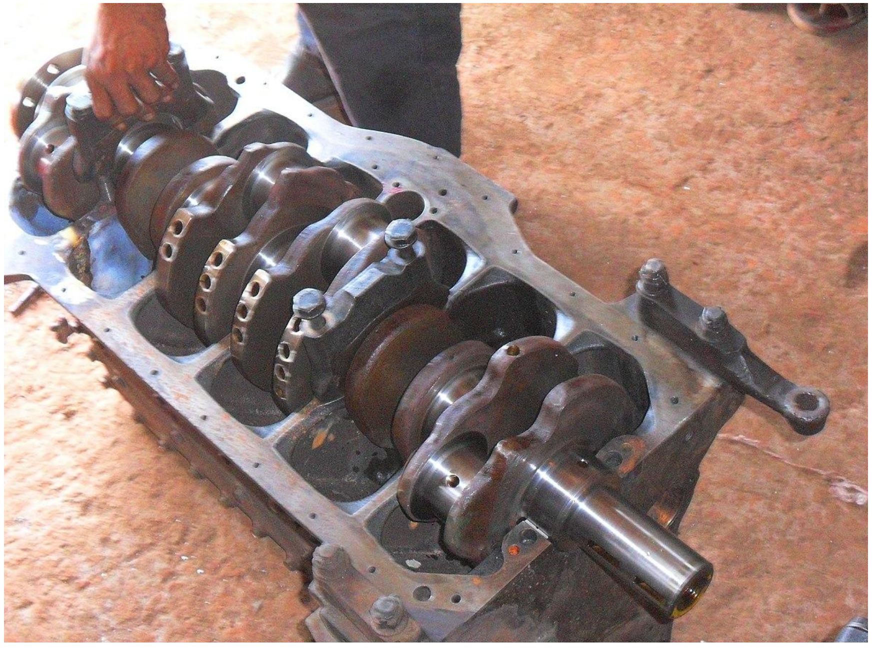
ME350 PUZZLE



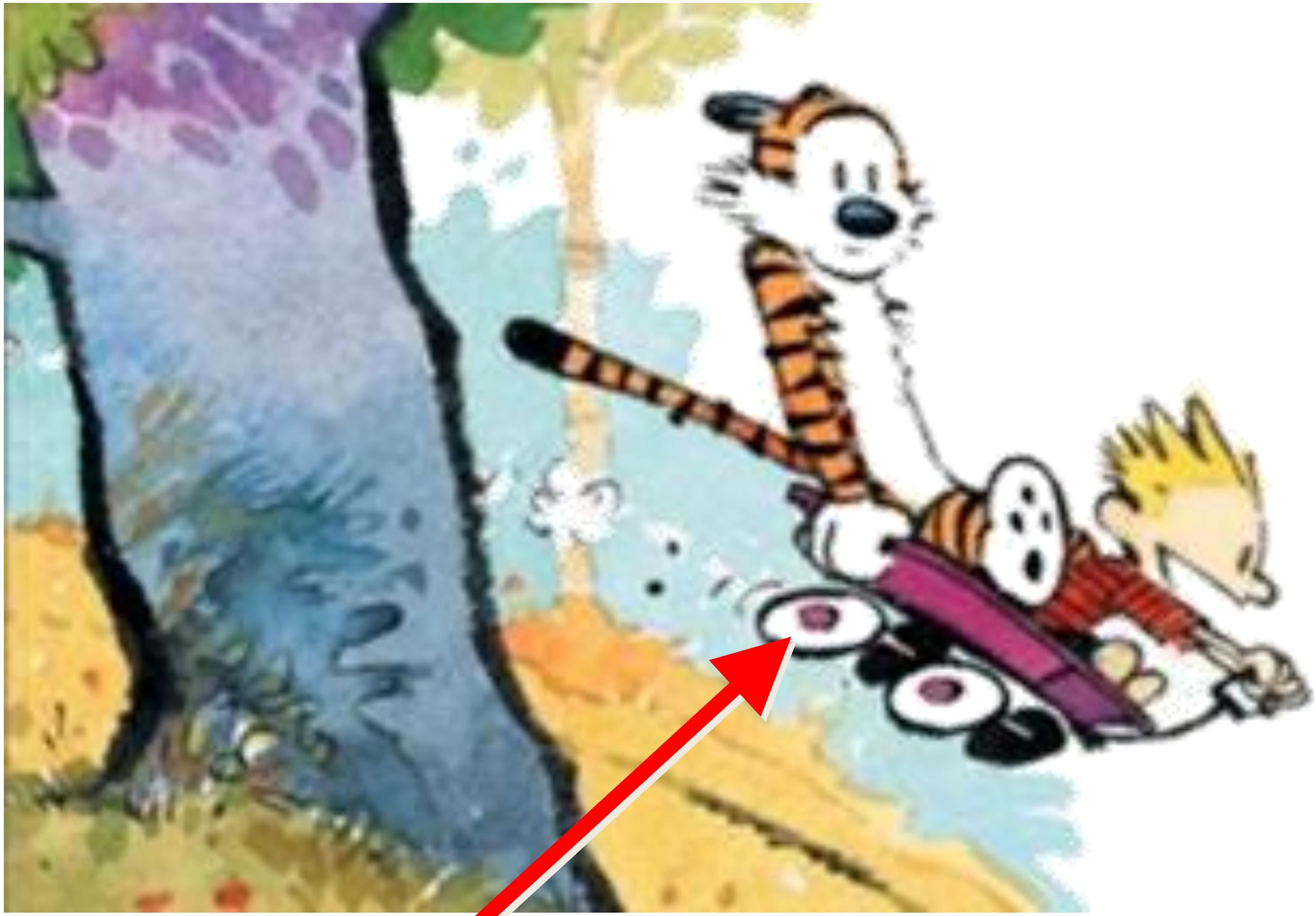
Which type of bearing (properly maintained) can be expected to last longer?

Crankshaft





Images: Kuba.Zahora, seedlesslime (CC)



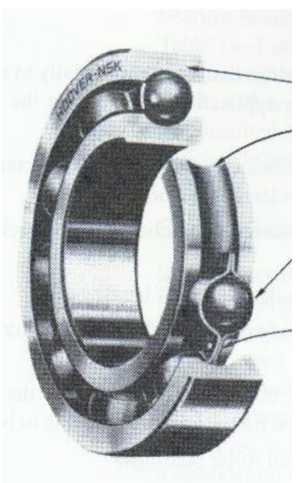
Bearings

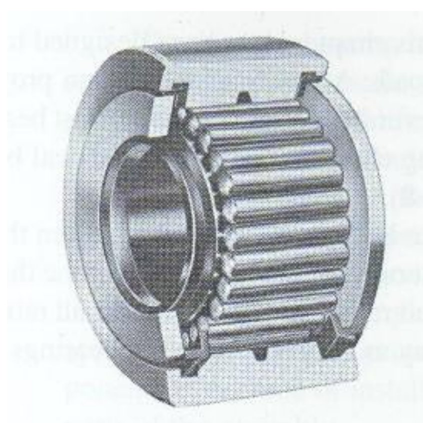
ME 350 Lecture
Mike Umbriac

Image: Bill Watterson

Sample Parts to Pass Around

- Ball Bearings
- Needle Roller Bearing
- Tapered Roller Bearing
- Bronze Bushing





Announcements / Reminders

- **During the next lab (Preliminary Testing of Project), take a video of your project's performance. Don't submit it yet, because you might take a video of better performance by the date of Final Testing.**

Reimbursement for ME350 Project Materials

- Scan your **itemized receipts** for **materials that you bought for the ME350 project**. The instructional team will allow purchases totaling **up to $100 per team** for any new materials purchased specifically for the ME350 project. If you have questions on what is allowed, send an email to mumbriac@umich.edu .

- ***Within 40 days of making the purchase***, fill out the Mechanical Engineering Travel & Expense Form. For purchases for the ME350 project, put "ME350" for the shortcode.

  Requests received after 40 days of making the purchase are at risk of not being reimbursed. If you think that you might make another purchase, you can submit an additional request at that time, if the total for all purchases is less than $100.

- The ME Admin office will review the request. If it gets approved, they will send a check to the requestor's home address.

Outline

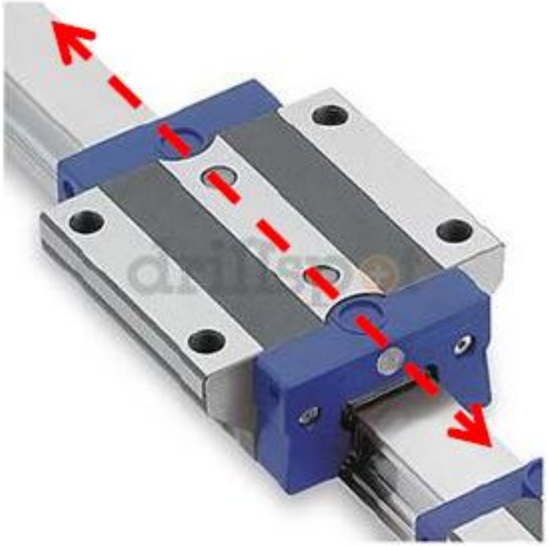
- **Review of Types and Application Examples**
- **Bearing Selection Calculations for Bearings with Radial loads only**
- **Bearing Selection Calculations for Reliability greater than 90%**
- **Bearing Failure Video**

- **Next lecture: Bearing Selection Calculations for combined radial and thrust loads; other modes of failure**

Purpose of Bearings

1. A bearing ALLOWS motion along certain directions.
   a) It minimizes friction or resistance to motion in these direction(s)
2. A bearing CONSTRAINS motion along the remaining directions
   a) It BEARS loads in these directions
   b) It minimizes any slop/backlash in these directions.

Revolute Joint



Prismatic Joint

Types of Loads on Bearings

Thrust Loads
(Axial Loads)

Radial Loads

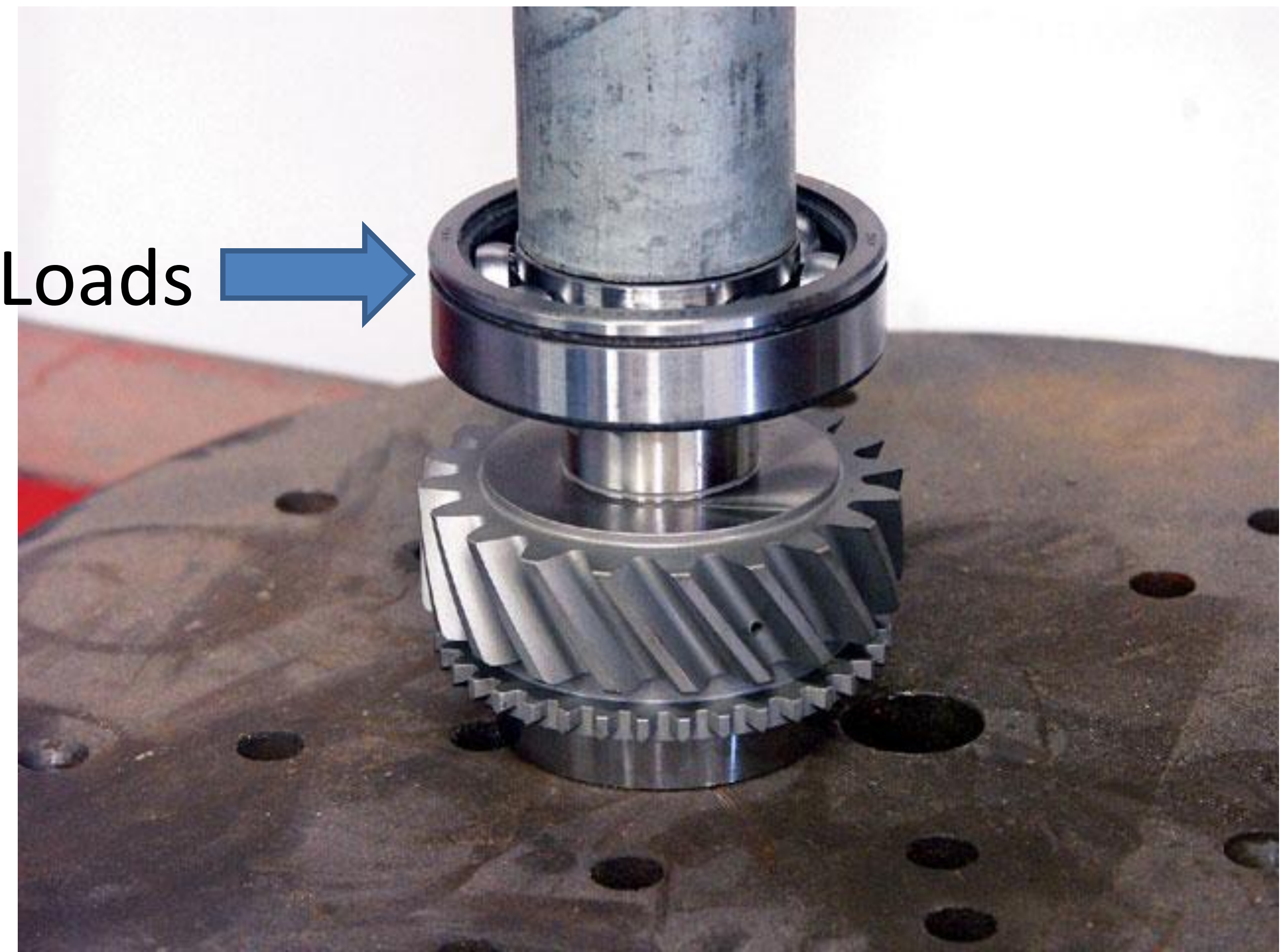
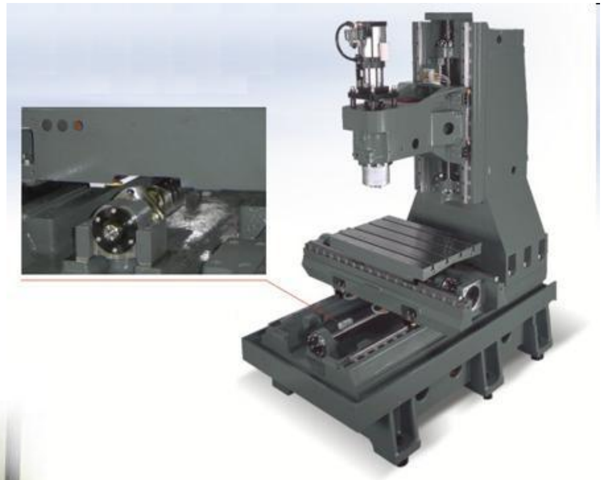


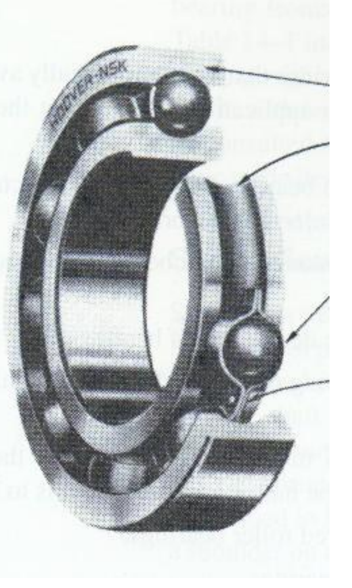
Image: Carcraft.com

Example Bearing Applications

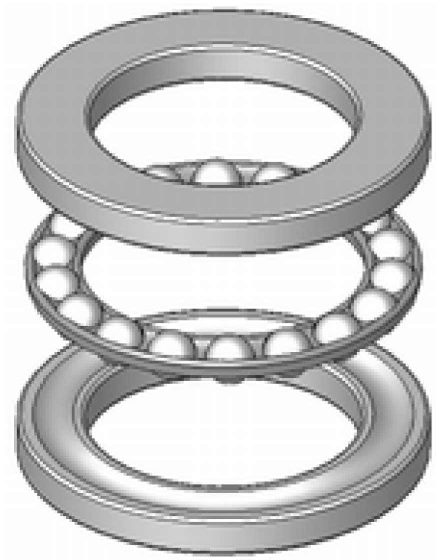


Types of Bearings

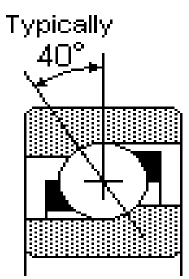
- Rolling Contact Bearings
  - Ball Bearings
    - Deep Groove
    - Angular Contact
    - Thrust
  - Roller Bearings
    - Cylindrical
    - Tapered
    - Needle
    - Spherical
  - Linear Bearings
- Sliding Contact Bearings
  - Journal (Plain) Bearings



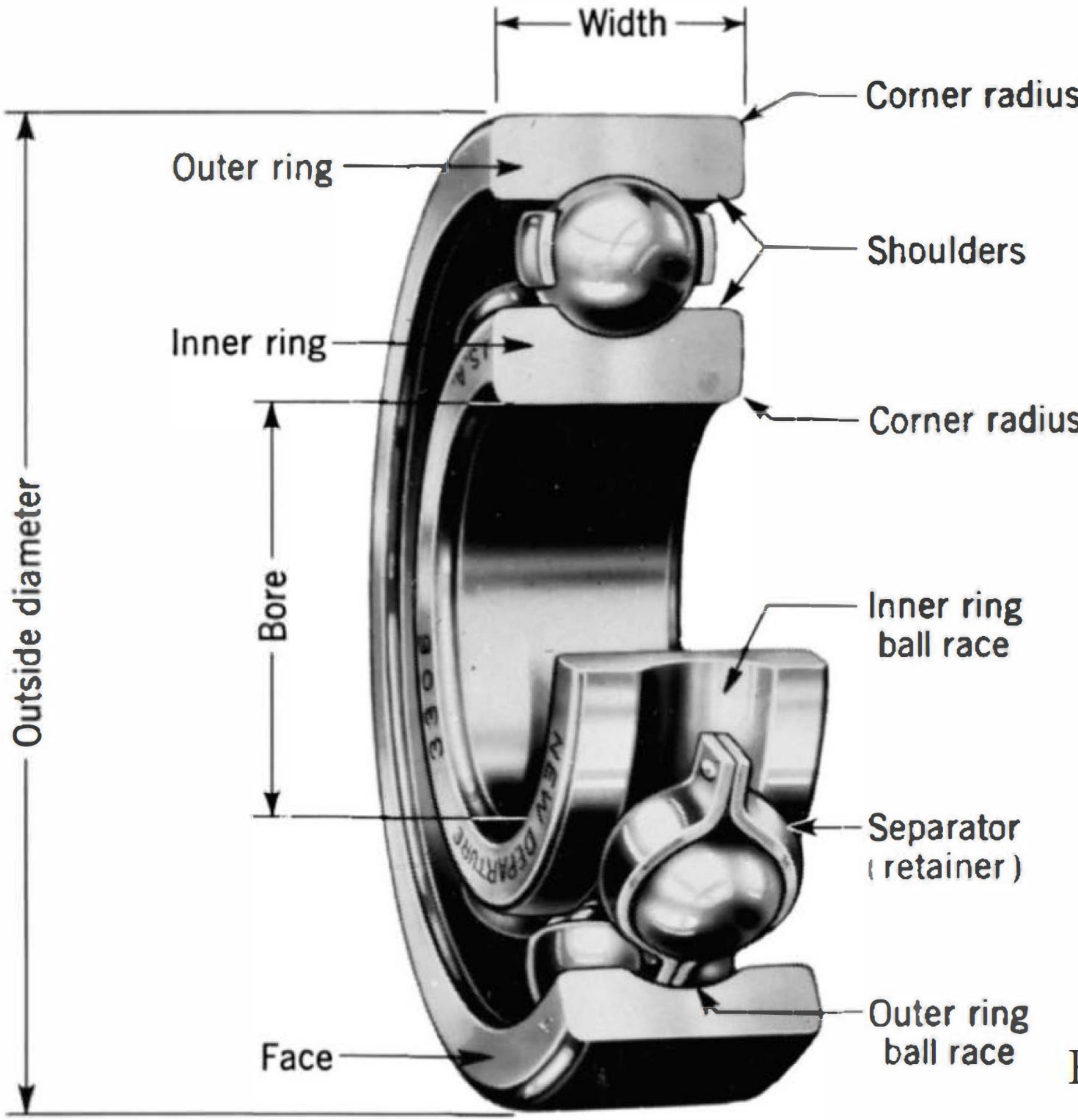
Ball Bearings



- Deep Groove Bearings
- Angular Contact Bearings
- Thrust Bearings



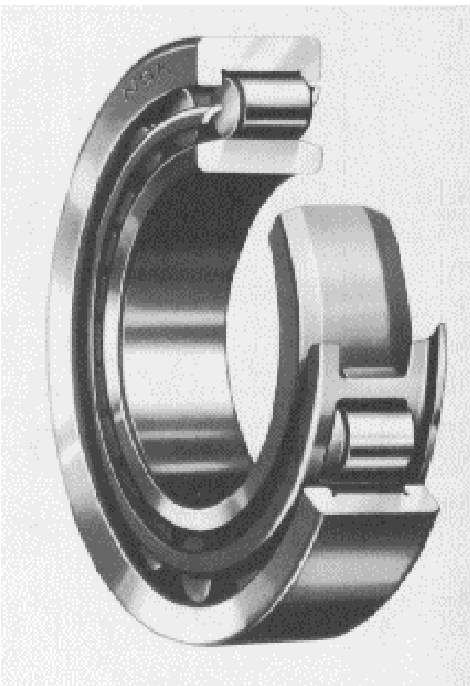
Nomenclature of a Ball Bearing



General Motors Corp. Used with permission, GM Media Archives.

Fig. 11–1

Roller Bearings



Cylindrical Roller

Tapered Roller

Spherical Roller

Alignment

- Catalogs will specify alignment requirements for specific bearings
- Typical maximum ranges for shaft slopes at bearing locations

Tapered roller	0.0005–0.0012 rad
Cylindrical roller	0.0008–0.0012 rad
Deep-groove ball	0.001–0.003 rad
Spherical ball	0.026–0.052 rad
Self-align ball	0.026–0.052 rad

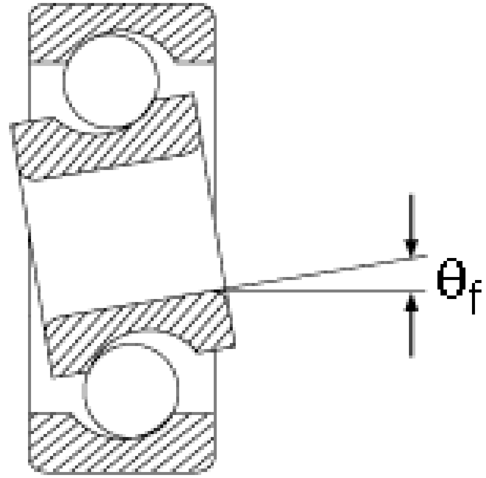


Image: dynaroll.com

Needle Roller Bearings

Images: ddb-tech.com, vxb.com

Type Characteristics

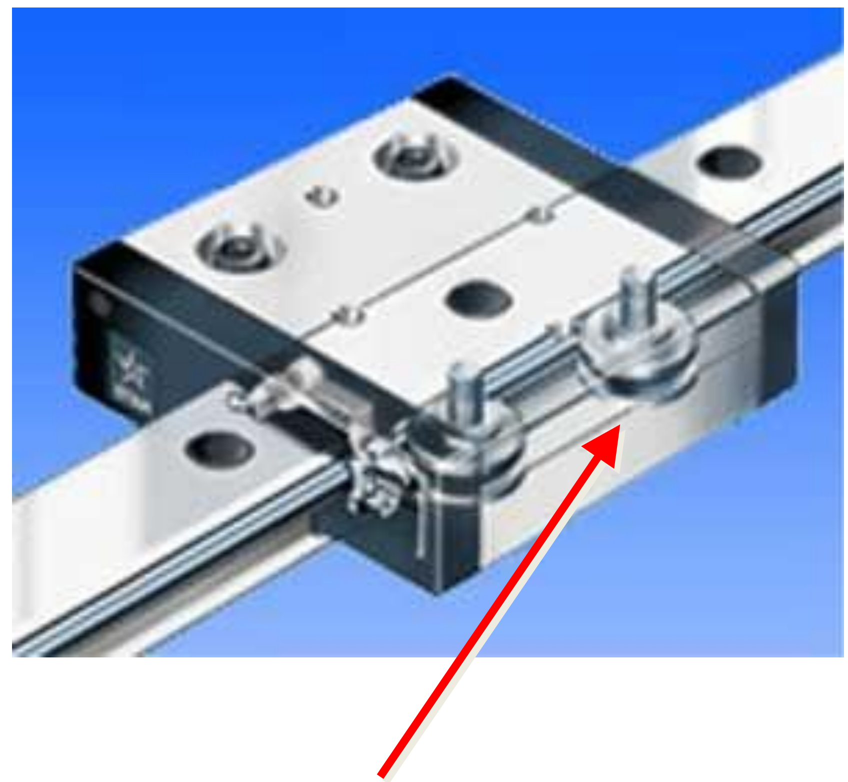
- Which type of bearing should I use?

Bearing type	Radial Load Capacity	Thrust load capacity	Misalignment capability
Single row, deep groove ball	Good	Fair	Fair
Double row, deep groove ball	Excellent	Good	Fair (not as good as SR)
Angular contact	Good	Excellent	Poor
Cylindrical roller	Excellent	Poor	Fair
Needle	Excellent	Poor	Poor
Spherical roller	Excellent	Fair	Excellent
Tapered roller	Excellent	Excellent	Poor

Cam Roller Bearings
(a.k.a. Cam Followers)







Images: globalspec.com, boschrexroth.com

Linear Bearings

Images: NSKamericas.com, monarchbearing.com

Pillow Block Bearings and Flange Bearings

Images: diytrade.com, thebigbearingstore.com

Bearing Selection

Factors to Consider when Selecting a Bearing from a Catalog

- **Type of load: radial, thrust, combination of both, steady or shock**
- **Magnitude of load**
- Rotation speed
- Shaft misalignment
- Diameter of both shaft and housing
- Packaging constraints
- **Desired life**
- **Reliability**
- Maintenance requirements
- Cost

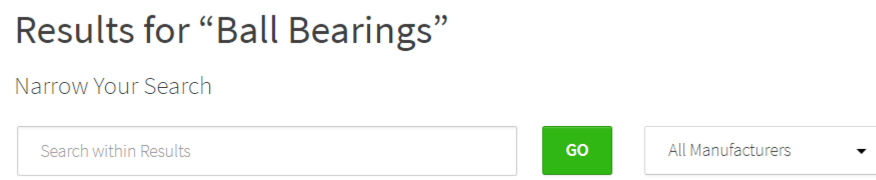
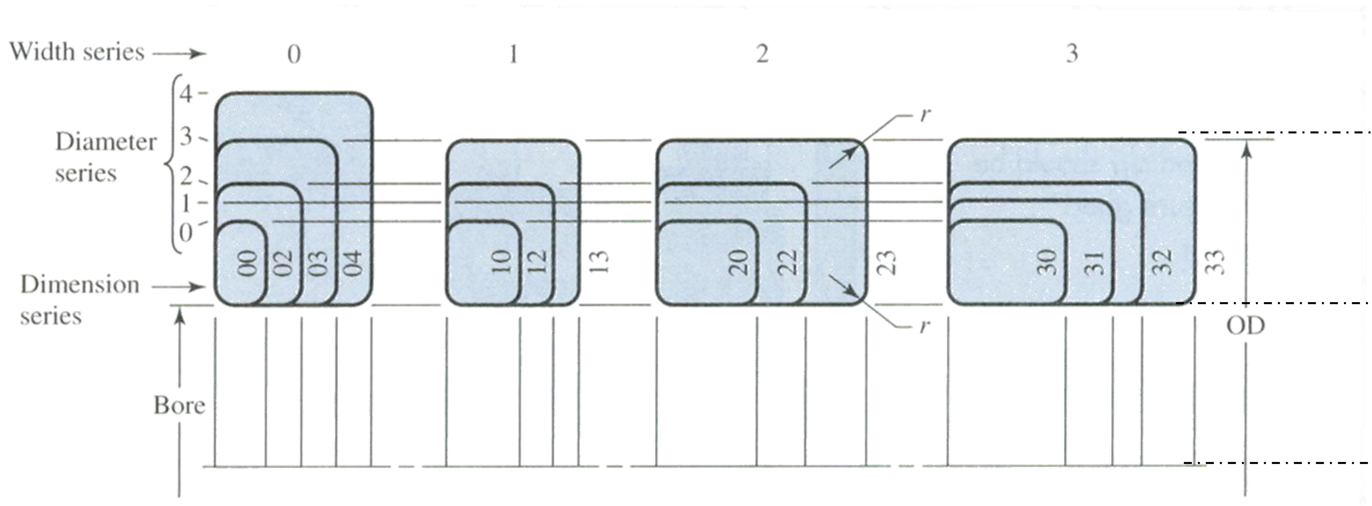
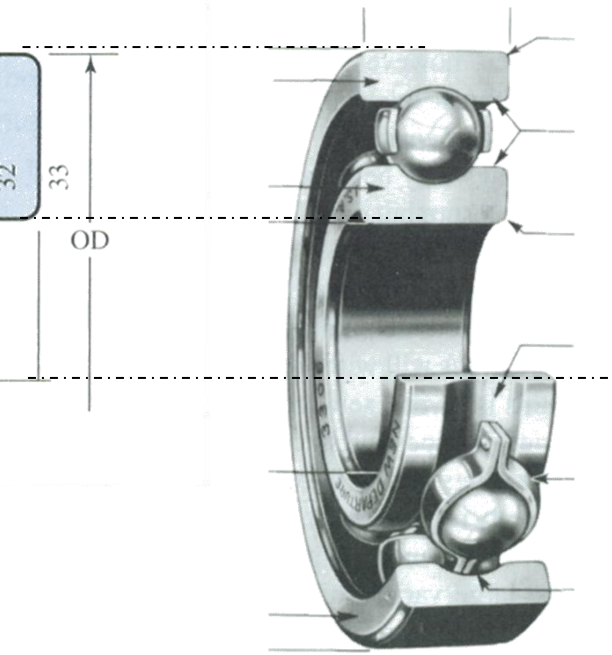


Image: Motion Industries website

ABMA Bearing Size Standards





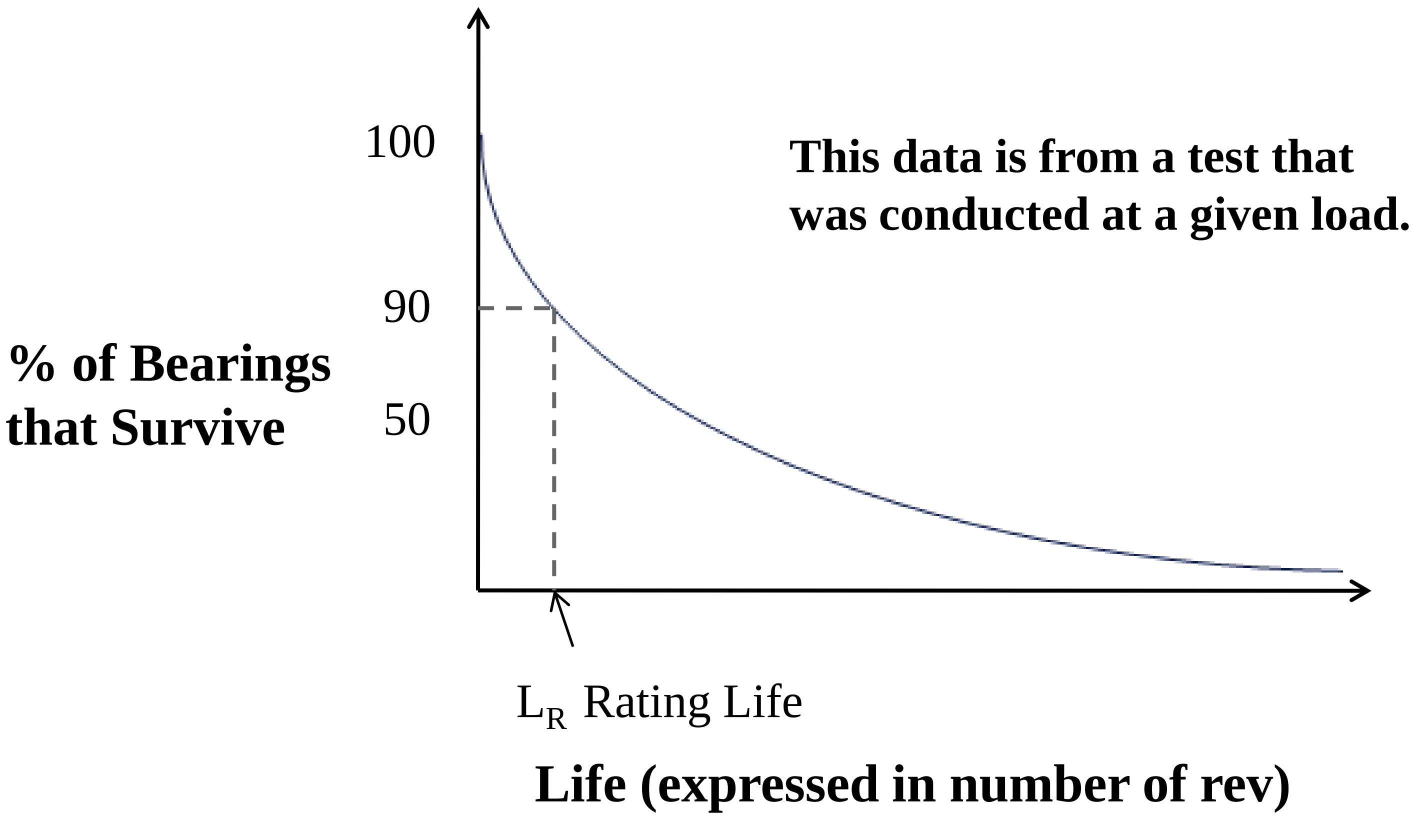
- Two digit series number
  - First digit designates the width series
  - Second digit designates the diameter series
- Specific dimensions are tabulated in catalogs under a specific series
- Determines the aspect ratio of the bearing
- This and the bore diameter determine the total size.

Note: Different bearing manufacturers may use different, but similar standards.

Failure due to dynamic loads (fatigue):

- The life, reliability, and bearing loading conditions are all related.
- Fatigue failures are typically due to high contact stresses.
- Signs of this type of failure are spalling or pitting in the area of contact in the bearing race.

Bearing Life Testing

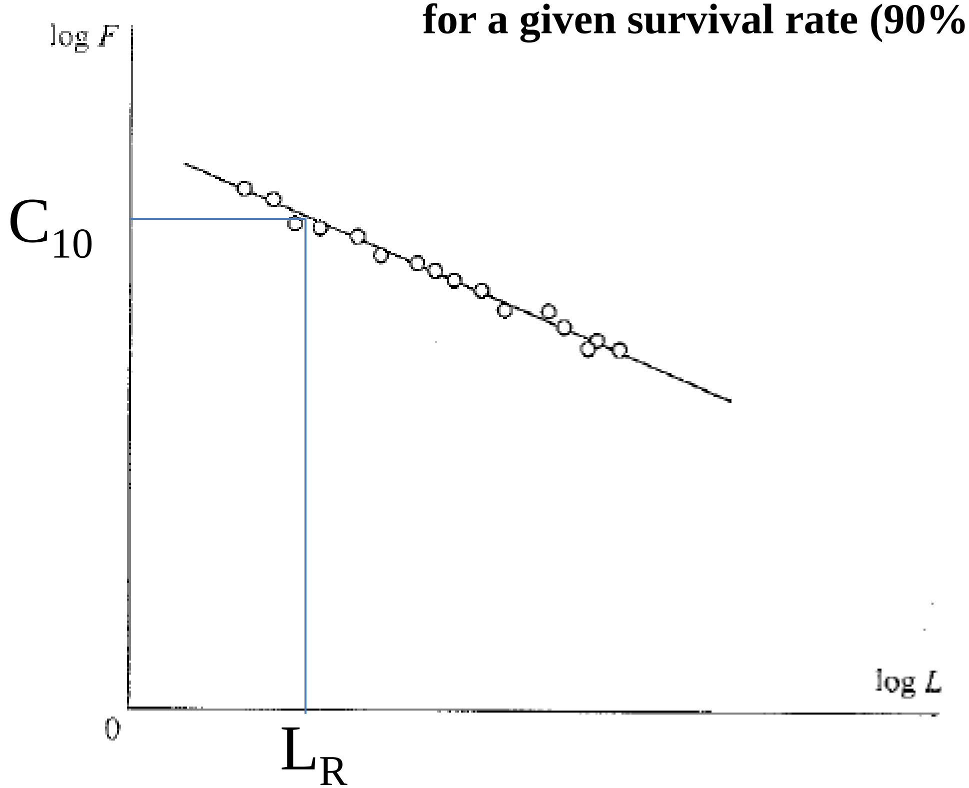


Bearing Life Testing

- **C_{10} is the radial load that gives L_R life = 10^6 revolutions.**
- In other words, it is the amount of radial load that the bearings can take for a million revolutions, before 10% of the bearings fail.

This data represents Load vs Life for a given survival rate (90%.)

Load (kN)



Life (expressed in number of rev)

Image: Shigley

Rating Life (L_R) and Catalog Load Rating (C_{10})

'Rating Life' L_R (sometimes called L_{10} life)

- **For most manufacturers, testing is done to 10^6 revolutions ($L_R=10^6$), and the C_{10} value below is measured from this testing.**
- **L_R or L_{10} represents 10% failure probability and 90% reliability.**

'Catalog Load Rating' C_{10}

- **The radial load that causes 10% of a group of bearings to fail (to show the first signs of fatigue), when tested at the manufacturer's L_R rating life.**
- **Also known as 'Basic Dynamic Load Rating'**

Representative Catalog Data for Ball Bearings (Table 11–2)

Dimensions and Load Ratings for Single-Row 02-Series Deep-Groove and Angular-Contact Ball Bearings

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder Diameter, mm		Load Ratings, kN			
						Deep Groove		Angular Contact	
				d_S	d_H	C_{10}	C_0	C_{10}	C_0
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6
45	85	19	1.0	52	77	33.2	18.6	35.8	21.2
50	90	20	1.0	56	82	35.1	19.6	37.7	22.8
55	100	21	1.5	63	90	43.6	25.0	46.2	28.5
60	110	22	1.5	70	99	47.5	28.0	55.9	35.5
65	120	23	1.5	74	109	55.9	34.0	63.7	41.5
70	125	24	1.5	79	114	61.8	37.5	68.9	45.5
75	130	25	1.5	86	119	66.3	40.5	71.5	49.0
80	140	26	2.0	93	127	70.2	45.0	80.6	55.0
85	150	28	2.0	99	136	83.2	53.0	90.4	63.0
90	160	30	2.0	104	146	95.6	62.0	106	73.5
95	170	32	2.0	110	156	108	69.5	121	85.0

Representative Catalog Data for Cylindrical Roller Bearings (Table 11–3)

Bore, mm	02-Series OD, mm	02-Series Width, mm	02-Series Load Rating, kN C_{10}	02-Series Load Rating, kN C_0	03-Series OD, mm	03-Series Width, mm	03-Series Load Rating, kN C_{10}	03-Series Load Rating, kN C_0
25	52	15	16.8	8.8	62	17	28.6	15.0
30	62	16	22.4	12.0	72	19	36.9	20.0
35	72	17	31.9	17.6	80	21	44.6	27.1
40	80	18	41.8	24.0	90	23	56.1	32.5
45	85	19	44.0	25.5	100	25	72.1	45.4
50	90	20	45.7	27.5	110	27	88.0	52.0
55	100	21	56.1	34.0	120	29	102	67.2
60	110	22	64.4	43.1	130	31	123	76.5
65	120	23	76.5	51.2	140	33	138	85.0
70	125	24	79.2	51.2	150	35	151	102
75	130	25	93.1	63.2	160	37	183	125
80	140	26	106	69.4	170	39	190	125
85	150	28	119	78.3	180	41	212	149
90	160	30	142	100	190	43	242	160
95	170	32	165	112	200	45	264	189
100	180	34	183	125	215	47	303	220
110	200	38	229	167	240	50	391	304
120	215	40	260	183	260	55	457	340
130	230	40	270	193	280	58	539	408
140	250	42	319	240	300	62	682	454
150	270	45	446	260	320	65	781	502

Bearing Life and Loading Conditions

The catalog load rating of a bearing can be calculated from:

$$\boxed{C_{10} = F_D \left(\frac{L_D}{L_R} \right)^{\frac{1}{a}}}$$

a = 3 for ball bearings
a = 10/3 for roller bearings

where: **C_{10} = load rating/catalog load (kN)**

F_D = desired load (kN)

L_D = desired lifetime (rev.)

L_R = rated lifetime (usually = 10^6 rev.) at the test load

You can then select bearings (from a catalog) that have a C_{10} higher than the C_{10} calculated above.

If L_D is specified as hours at a given speed *n*, convert to revolutions before you use the above equation.

NOTE: This equation is only valid for 90% reliability.

STRETCH!

Example Problem (Radial load only)

Select a 02-series deep-groove ball bearing to satisfy the following criteria, with 90% reliability:

The bearing needs to last for 3 years on a machine in a manufacturing plant that works 80 hours per week, 50 weeks per year.

The shaft on this machine will be turning at 1725 RPM.

The design load is 800 lb.

Representative Catalog Data for <u>Ball</u> Bearings (Table 11–2)

Dimensions and Load Ratings for Single-Row 02-Series Deep-Groove and Angular-Contact Ball Bearings

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder Diameter, mm		Load Ratings, kN			
						Deep Groove		Angular Contact	
				d_S	d_H	C_{10}	C_0	C_{10}	C_0
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6
45	85	19	1.0	52	77	33.2	18.6	35.8	21.2
50	90	20	1.0	56	82	35.1	19.6	37.7	22.8
55	100	21	1.5	63	90	43.6	25.0	46.2	28.5
60	110	22	1.5	70	99	47.5	28.0	55.9	35.5
65	120	23	1.5	74	109	55.9	34.0	63.7	41.5
70	125	24	1.5	79	114	61.8	37.5	68.9	45.5
75	130	25	1.5	86	119	66.3	40.5	71.5	49.0
80	140	26	2.0	93	127	70.2	45.0	80.6	55.0
85	150	28	2.0	99	136	83.2	53.0	90.4	63.0
90	160	30	2.0	104	146	95.6	62.0	106	73.5
95	170	32	2.0	110	156	108	69.5	121	85.0

Recommendations for Bearing Design

TYPE OF APPLICATION	LIFE, kh
Instruments and apparatus for infrequent use	Up to 0.5
Aircraft engines	0.5–2
Machines for short or intermittent operation where service interruption is of minor importance	4–8
Machines for intermittent service where reliable operation is of great importance	8–14
Machines for 8-h service which are not always fully utilized	14–20
Machines for 8-h service which are fully utilized	20–30
Machines for continuous 24-h service	50–60
Machines for continuous 24-h service where reliability is of extreme importance	100–200

Juvinall and Marshek, *Fundamentals of Machine Component Design*

Recommendations for Bearing Design

Recommended Design Life for Bearings

Application	*Design Life* L_{10}, *h*
Domestic appliances	1 000–2 000
Aircraft engines	1 000–4 000
Automotive	1 500–5 000
Agricultural equipment	3 000–6 000
Elevators, industrial fans, multipurpose gearing	8 000–15 000
Electric motors, industrial blowers, general industrial machines	20 000–30 000
Pumps and compressors	40 000–60 000
Critical equipment in continuous 24-h operation	100 000–200 000

Source: Eugene A. Avallone and Theodore Baumeister III, eds. *Marks' Standard Handbook for Mechanical Engineers*, 9th ed. New York: McGraw-Hill Book Company, 1986.

Bearing Selection for Reliabilities higher than 90%

Suppose a reliability higher than 90% is desired.

- It is often convenient to define a dimensionless *multiple of rating life:*

$$\boxed{x_D = L_D/L_R}$$

where

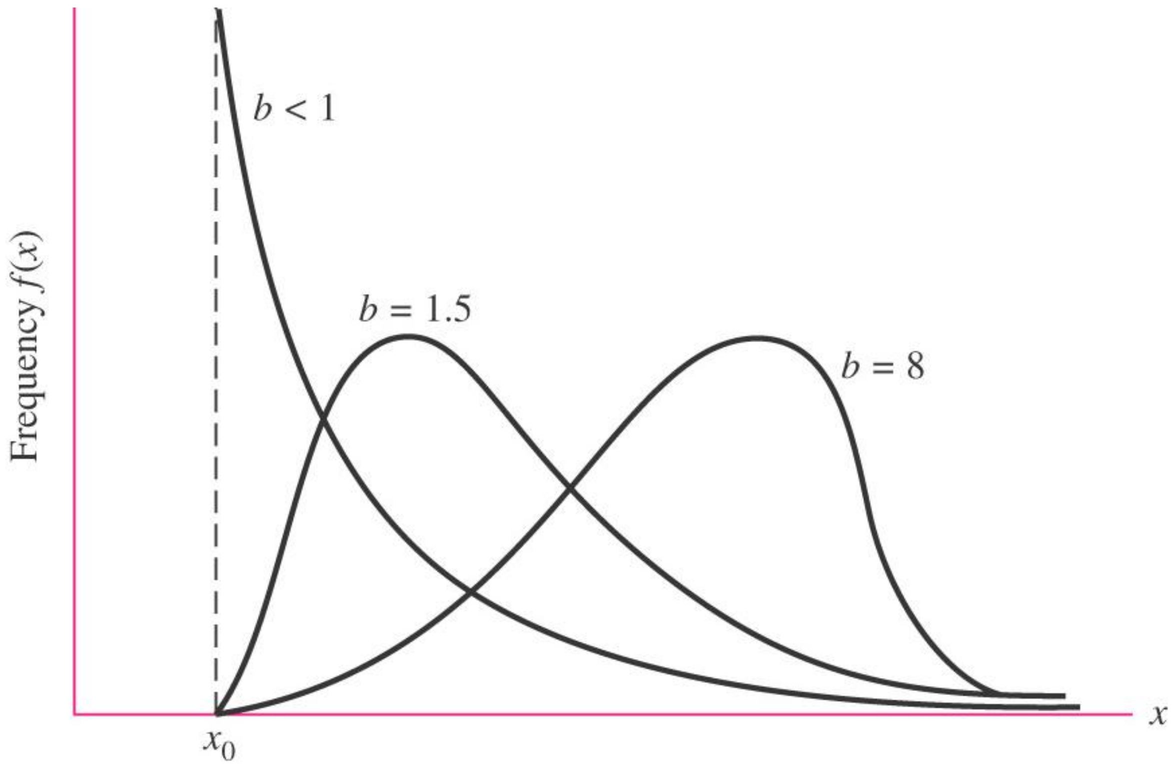
L_D = actual lifetime (rev.)

L_R = rated lifetime at test load (usually 10^6 rev.)

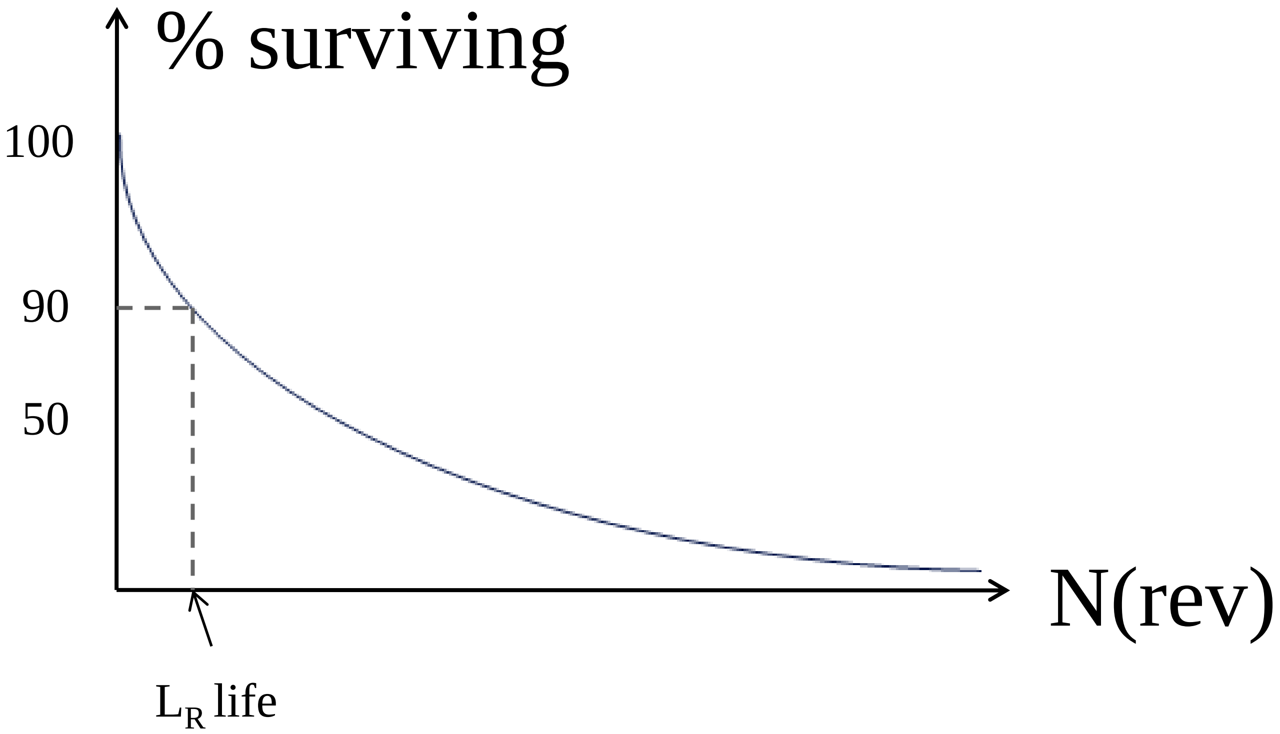
- The needed catalog load rating, C_{10}, can be calculated using the **Weibull distribution.**

Side note: Weibull Distribution

- The *Weibull distribution* is a flexible, asymmetrical probability distribution with different values for mean and median.
- It contains within it a good approximation of the normal distribution and an exact representation of the exponential distribution.
- It is widely used to represent laboratory and field service data.



Bearing Life Testing



Side note: Weibull Distribution

the three parameters are

$$x_0 = \text{minimum, guaranteed, value of } x$$
$$\theta = \text{a characteristic or scale value } (\theta \geq x_0)$$
$$b = \text{a shape parameter } (b > 0)$$

- The *characteristic variate* θ represents a value of x below which lie 63.2% of the observations.
- The *shape parameter* b controls the skewness of the distribution.
- A good approximation to the normal distribution is obtained when $3.3 < b < 3.5$
- The distribution is exponential when $b = 1$

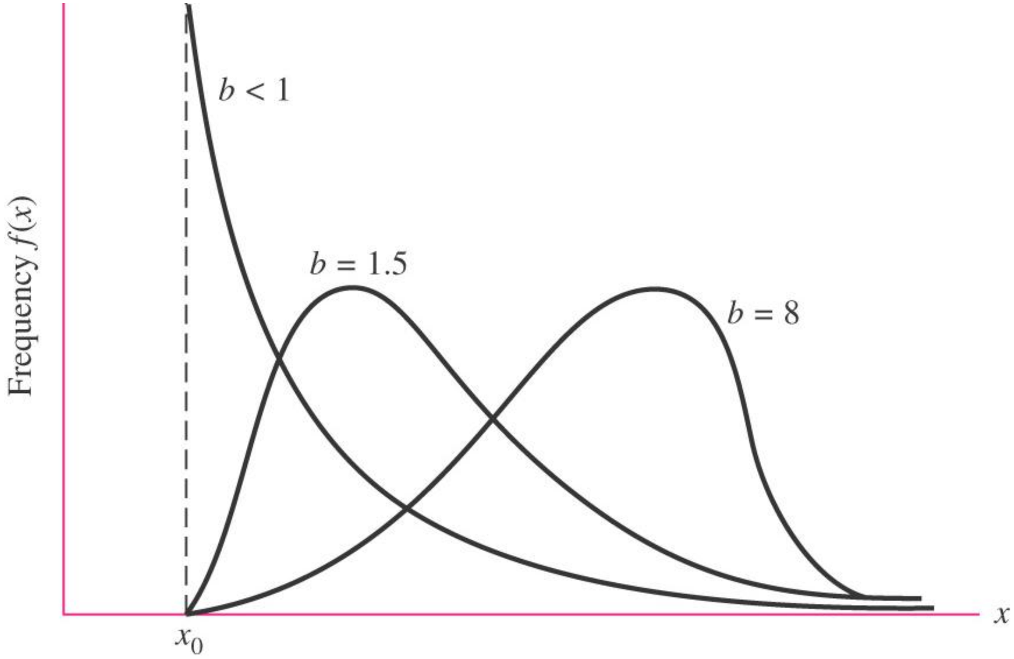


Fig. 20–9

Bearing Selection for Reliabilities higher than 90%

Here is the equation we will use, derived from the Weibull distribution:

$$\boxed{C_{10} \doteq a_f F_D \left[\frac{x_D}{x_0 + (\theta - x_0)(1 - R_D)^{1/b}} \right]^{1/a} \qquad R \geq 0.90} \qquad (11\text{–}7)$$

(Refer to the application factor a_f and the Weibull parameters x_0, θ, and b on the following slides.)

Recommended Load Application Factors, a_f (Table 11–5)

Type of Application	Load Factor
Precision gearing	1.0–1.1
Commercial gearing	1.1–1.3
Applications with poor bearing seals	1.2
Machinery with no impact	1.0–1.2
Machinery with light impact	1.2–1.5
Machinery with moderate impact	1.5–3.0

- The **application factor a_f** serves as a factor of safety. It is multiplied by the design load to take into account overload, dynamic loading, and uncertainty.

Weibull Parameters for Bearings

- The Weibull parameters x_0, θ, and b are usually provided by the manufacturer's catalog.
- Typical values of Weibull parameters are shown below.
- Manufacturer 1 parameters are common for tapered roller bearings
- Manufacturer 2 parameters are common for ball and straight roller bearings

Manufacturer	Rating Life, Revolutions	Weibull Parameters Rating Lives x_0	θ	b
1	$90(10^6)$	0	4.48	1.5
2	$1(10^6)$	0.02	4.459	1.483

Example Problem: Reliability of 99%

A cylindrical roller bearing is to be used in an application where the design load is 550 lb. Because the machine has a possibility of experiencing light impact load from time to time, use an application factor of 1.2. The machine will operate at 350 rpm. A design life of 30,000 hours and a reliability of 99% are desired. Find the catalog load rating to be met or exceeded when selecting this bearing from a catalog.

Useful equations:

$$x_D = L_D / L_R$$

$$C_{10} \doteq a_f F_D \left[\frac{x_D}{x_0 + (\theta - x_0)(1 - R_D)^{1/b}} \right]^{1/a} \qquad R \geq 0.90 \qquad (11\text{–}7)$$

Video:
Epic Universal Motor Death (Bearing Failure)

http://www.youtube.com/watch?v=0usO4r3rlCQ